

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. INTRODUCTION

The coding and solution phase details the implementation of core features, database-free stateless execution, and the rigorous troubleshooting undertaken to resolve real-world networking, DNS, and API rate-limit errors during the development of ComicCraft.

3. CORE CODE IMPLEMENTATIONS

A. FastAPI Main Routing (`routes.py`)

The backend uses FastAPI to handle user form submissions asynchronously and manage the sequential AI generation pipeline:

Python

```
@router.post("/generate", response_class=HTMLResponse)
async def generate_comic(request: Request, prompt: str =
Form(...), character_name: str = Form(...)):
    # Step 1: Generate Outline via Gemini Flash
    outline = generate_outline(prompt, character_name)
    # Step 2: Generate Full Story via Gemini Pro
    full_story = generate_story(outline)
    # Step 3: Generate Panel Images via Diffusion APIs with
    Fallbacks
    images = [generate_image(p["image_prompt"]) for p in
outline]
    # Step 4: Build Layout & Export PDF
    layout = build_comic_layout(images, full_story, outline)
    pdf_path = export_to_pdf(layout)
    return templates.TemplateResponse("comic_preview.html",
{"request": request, "layout": layout, "pdf_path":
```

```
pdf_path}))
```

B. Dual-Engine AI Image Generator (image_generator.py)

To prevent system crashes during API blocks (HTTP 402/503), the application implements a robust dual-engine fallback structure combining Pollinations AI and Hugging Face Stable Diffusion, backed by a Python PIL comic-caption text generator:

Python

```
def generate_image(prompt, filename=None):
    # Engine 1: Pollinations Multi-Node Bypass
    try:
        url =
f"https://image.pollinations.ai/prompt/{quote(prompt)}"
        # Opens URL and saves image...
        return path
    except Exception:
        pass

    # Engine 2: Hugging Face Stable Diffusion API
    try:
        hf_token = os.getenv("HF_API_KEY")
        # Calls Hugging Face Inference API...
        return path
    except Exception:
        # Ultimate Fallback: Python PIL Professional Comic
Caption Panel
        img = Image.new('RGB', (512, 512), color=(255, 215,
0))

        draw = ImageDraw.Draw(img)
        draw.text((30, 30), "MEANWHILE...", fill="black")
        img.save(path)
```

return path

4. TECHNICAL CHALLENGES & SOLUTIONS LOGGED DURING DEVELOPMENT

- **Error 1: Long Path Name OS Error [WinError 206] during pip install**
 - *Cause:* Installing heavy machine learning libraries (torch, diffusers) exceeded Windows default MAX_PATH character limits (260 characters).
 - *Solution:* Enabled long path support in the Windows Registry (Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\FileSystem -> LongPathsEnabled set from 0 to 1).
- **Error 2: PowerShell Execution Policy Restriction (UnauthorizedAccess)**
 - *Cause:* Local PowerShell security settings blocked the activation script of the virtual environment.
 - *Solution:* Executed Set-ExecutionPolicy RemoteSigned - Scope CurrentUser in the terminal to authorize script execution.
- **Error 3: Missing Module Dependencies (uvicorn, fastapi, jinja2, google)**
 - *Cause:* Incremental module requirements during package resolution.
 - *Solution:* Configured a comprehensive requirements.txt and systematically resolved dependencies using targeted pip install commands inside the activated virtual environment (env\Scripts\activate).
- **Error 4: DNS Resolution Failure ([Errno 11001] getaddrinfo failed)**
 - *Cause:* Local ISP network restrictions blocking direct calls to external Hugging Face model endpoints.
 - *Solution:* Implemented explicit proxy handlers (urllib.request.ProxyHandler({})) and configured robust multi-node fallback routes.